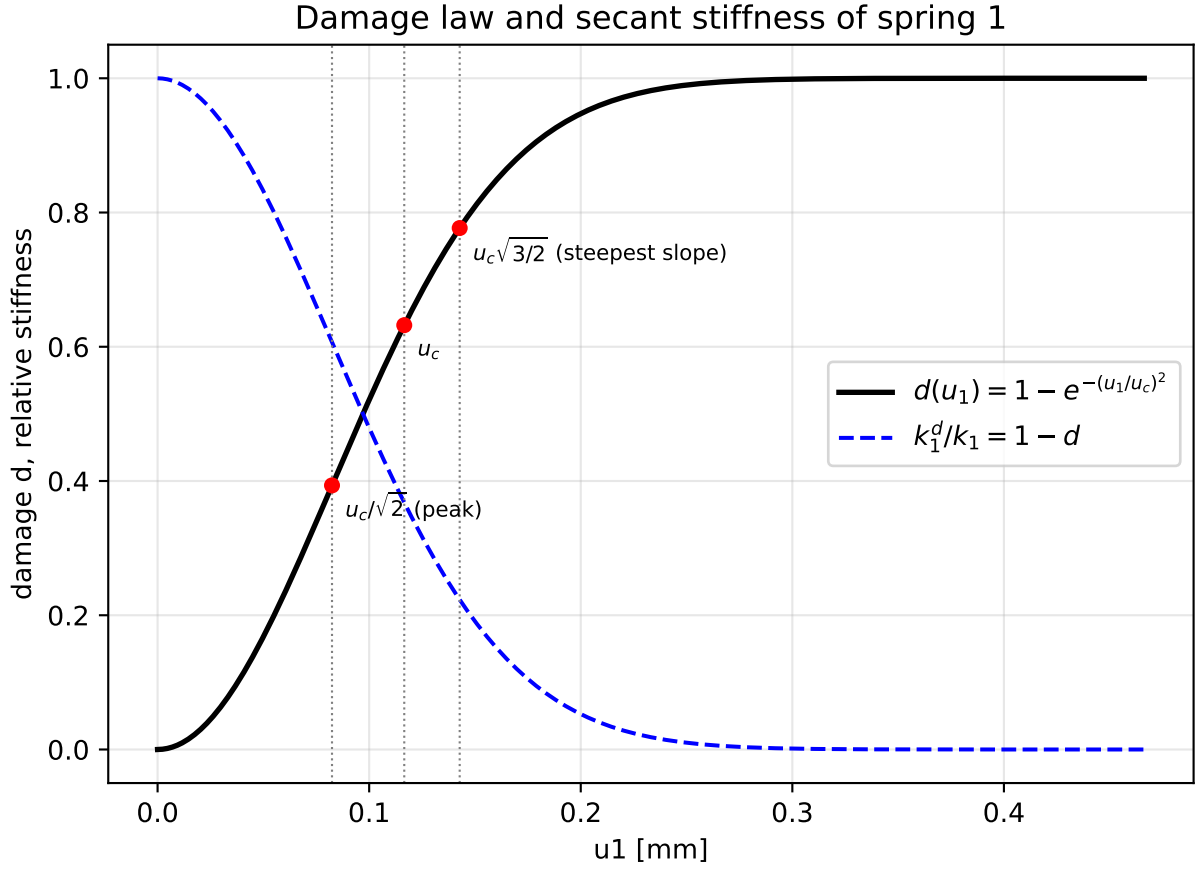
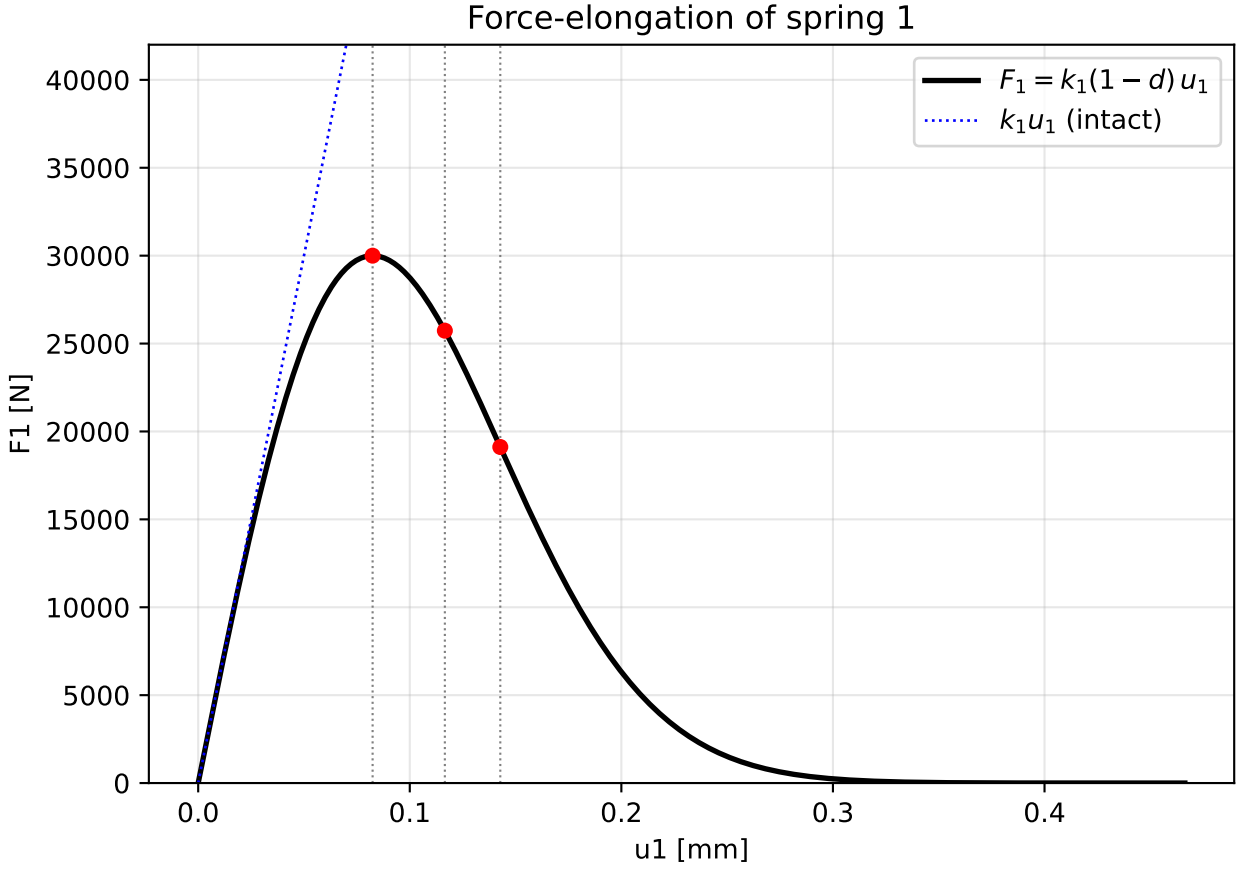

Damage law and secant stiffness of spring 1
damage d, relative stiffness
u1 [mm]
$d(u_1) = 1 - e^{-(u_1/u_c)^2}$
$k_1^d/k_1 = 1 - d$
$u_c\sqrt{3/2}$ (steepest slope)
u_c
$u_c/\sqrt{2}$ (peak)


Force-elongation of spring 1
F1 [N]
u1 [mm]
$F_1 = k_1(1-d)\,u_1$
$k_1 u_1$ (intact)